



Oncor Southern DFW Load Interconnection and General Grid Strengthening Project – ERCOT Independent Review (EIR) Status Update

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RPG Meeting
March 17, 2026

Introduction

- Oncor submitted the Southern Dallas Fort Worth (DFW) Load Interconnection and General Grid Strengthening Project for Regional Planning Group (RPG) review in February 2025
 - This Tier 1 project is estimated to cost approximately \$1.219 billion and will require a Certificate of Convenience and Necessity (CCN) filing
 - Expected in-service date (ISD) is December 2028
 - Addresses the thermal overloads and voltage violations due to proposed load additions in the Southern DFW in the North Central and East Weather Zones
- Oncor presented project overview and ERCOT presented Scope at the April 2025 RPG meeting
 - <https://www.ercot.com/calendar/04292025-RPG-Meeting>
- ERCOT presented status updates at the July and August 2025, and January 2026 RPG meetings
 - <https://www.ercot.com/calendar/07292025-RPG-Meeting>
 - <https://www.ercot.com/calendar/08262025-RPG-Meeting>
 - <https://www.ercot.com/calendar/01162026-RPG-Meeting>
- This project is currently under ERCOT Independent Review (EIR)

Study Assumptions

- Final 2024 Regional Transmission Planning (RTP) 2029 summer peak load case was used as the start case
- Transmission updates
 - List of new transmission projects added to the base case are in [Appendix A.1](#)
 - List of placeholder projects from the 2024 RTP that were removed from the study base case are in [Appendix A.2](#)
 - Additional updates based on feedback from Oncor were also applied to the study base case
- Generation updates
 - List of new generation added to the study base case are in [Appendix B](#)
 - All new generation were dispatched consistent with the 2024 RTP methodology
 - GenHubs in East and North Central Weather Zones were opened (turned off) to balance power
- Load updates
 - All 2024 RTP Officer Letter Loads (OLLs) (approximately 10.2 GW) in the study area were removed from the study base case
 - Large Loads including OLLs (approximately 4.0 GW) relevant to the project, provided by Oncor, in the study area were added to the study base case
 - Additional load updates (approximately 50 MW) in the study area were applied based on feedback from Brazos Electric
- Reserve was maintained consistent with the 2024 RTP

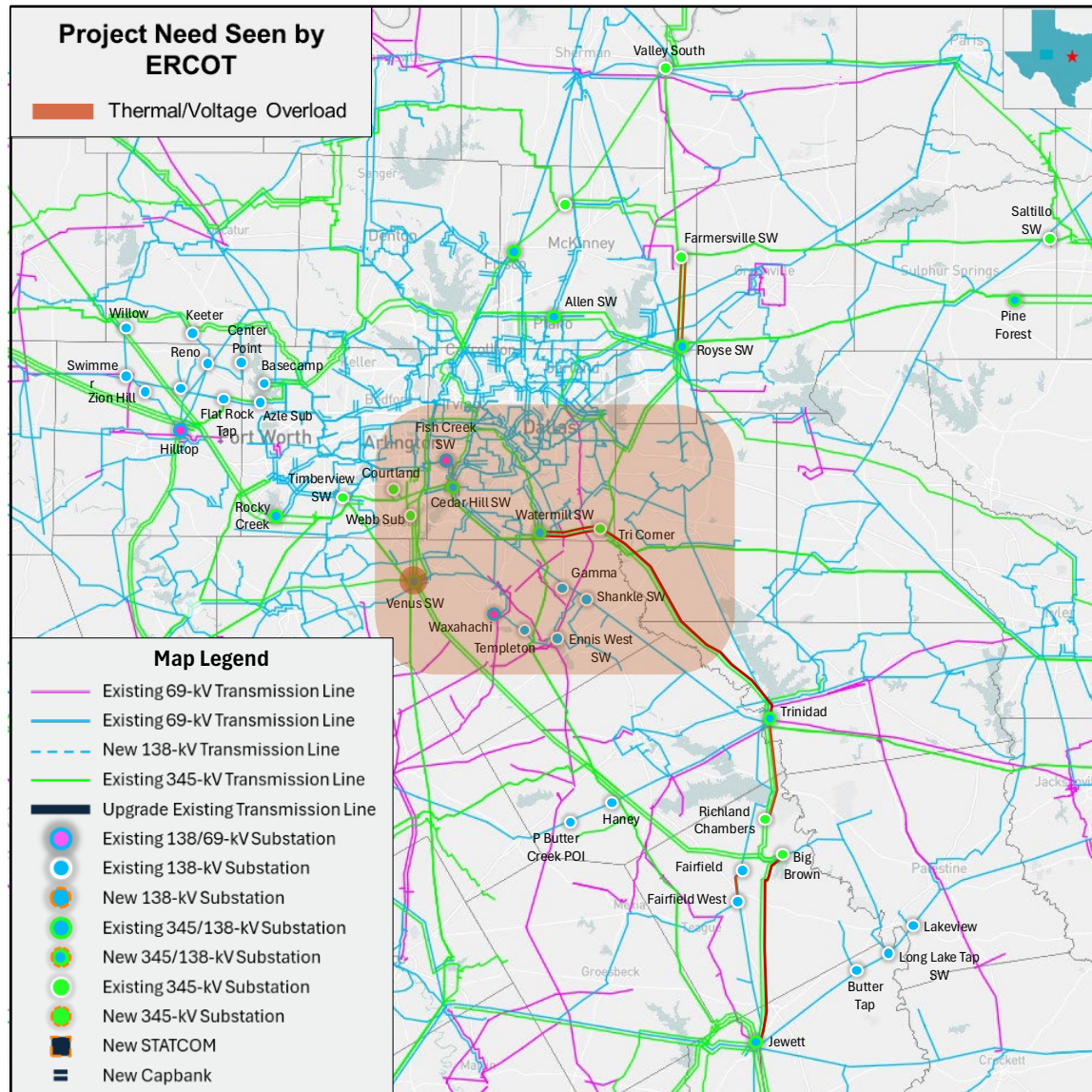
Results of Reliability Assessment – Need Analysis

- ERCOT conducted steady-state load flow analysis for the study base case according to the NERC Reliability Standard TPL-001-5.1 and ERCOT Planning Criteria to identify the project need

Contingency Category	Voltage Violations	Thermal Overloads	Unsolved Power Flow
N-0 (P0)	None	1	None
N-1 (P1, P2-1, P7)	6	14	None
G-1*+N-1 (P3)	8	10	None
X-1*+N-1 (P6-2)	30	23	3
Total	44	48	3

* See [Appendix C](#) for list of G-1 generators and X-1 transformers tested

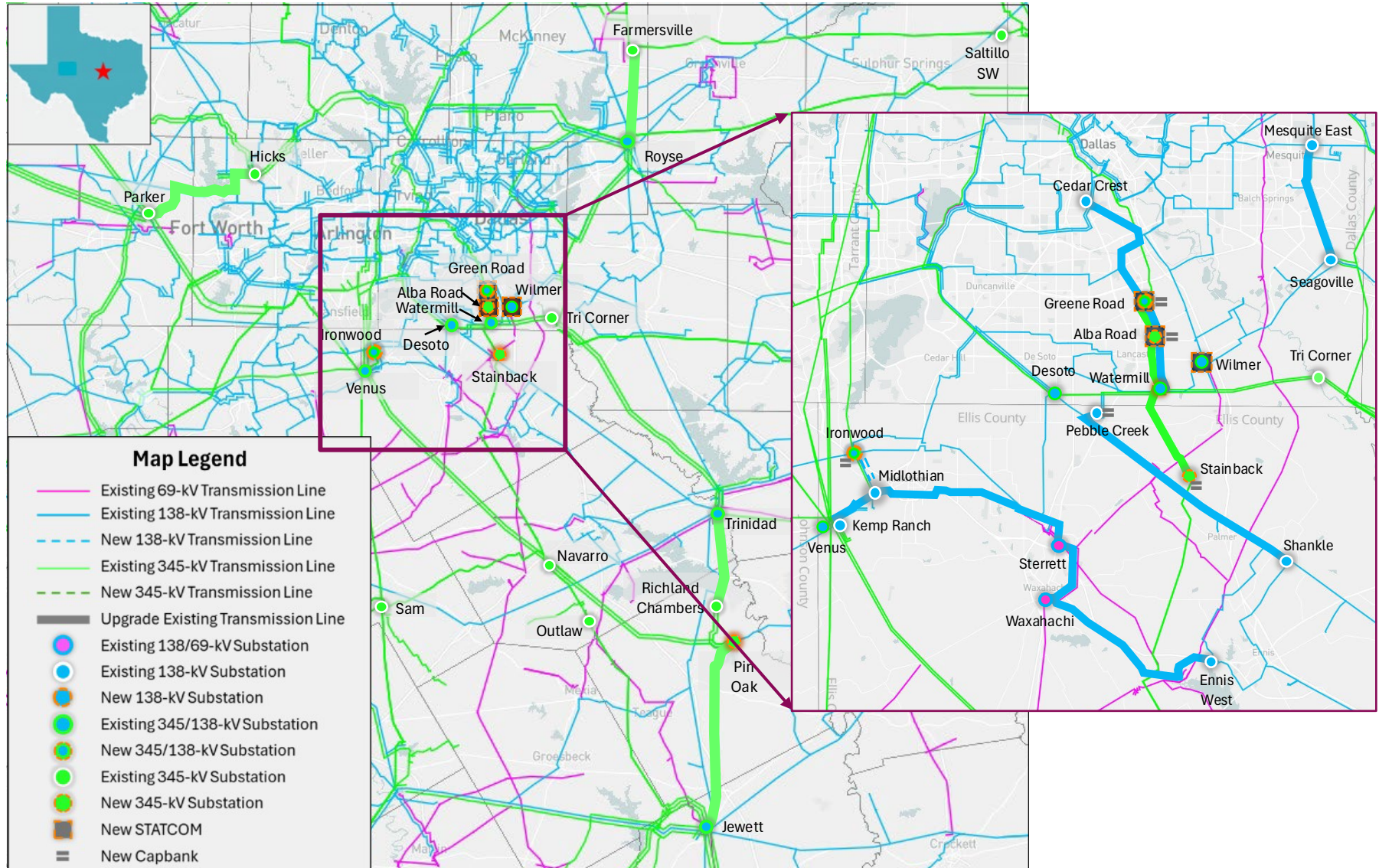
Map with All Violations Seen by ERCOT



Option 1 – Oncor Proposed Project Components Summary

- Establish four new switch stations
 - Greene Road 345/138-kV Switch
 - Ironwood 345/138-kV Switch
 - Stainback 345-kV Switch
 - Alba Road 345-kV Switch
- Rebuild one existing 345-kV switch station
- Construct four new 138-kV lines, approximately 8.0 circuit miles
- Upgrade existing lines
 - Existing 345-kV lines, approximately 169.8 circuit miles
 - Existing 138-kV lines, Approximately 73.1 circuit miles
- Nine reactive devices
 - Three ± 250 MVA 345-kV Static Synchronous Compensator (STATCOMs)
 - Three 345-kV capacitor banks
 - Three 138-kV capacitor banks
- Full project details in [Appendix D](#)

Map of Option 1 – Oncor Proposed Project



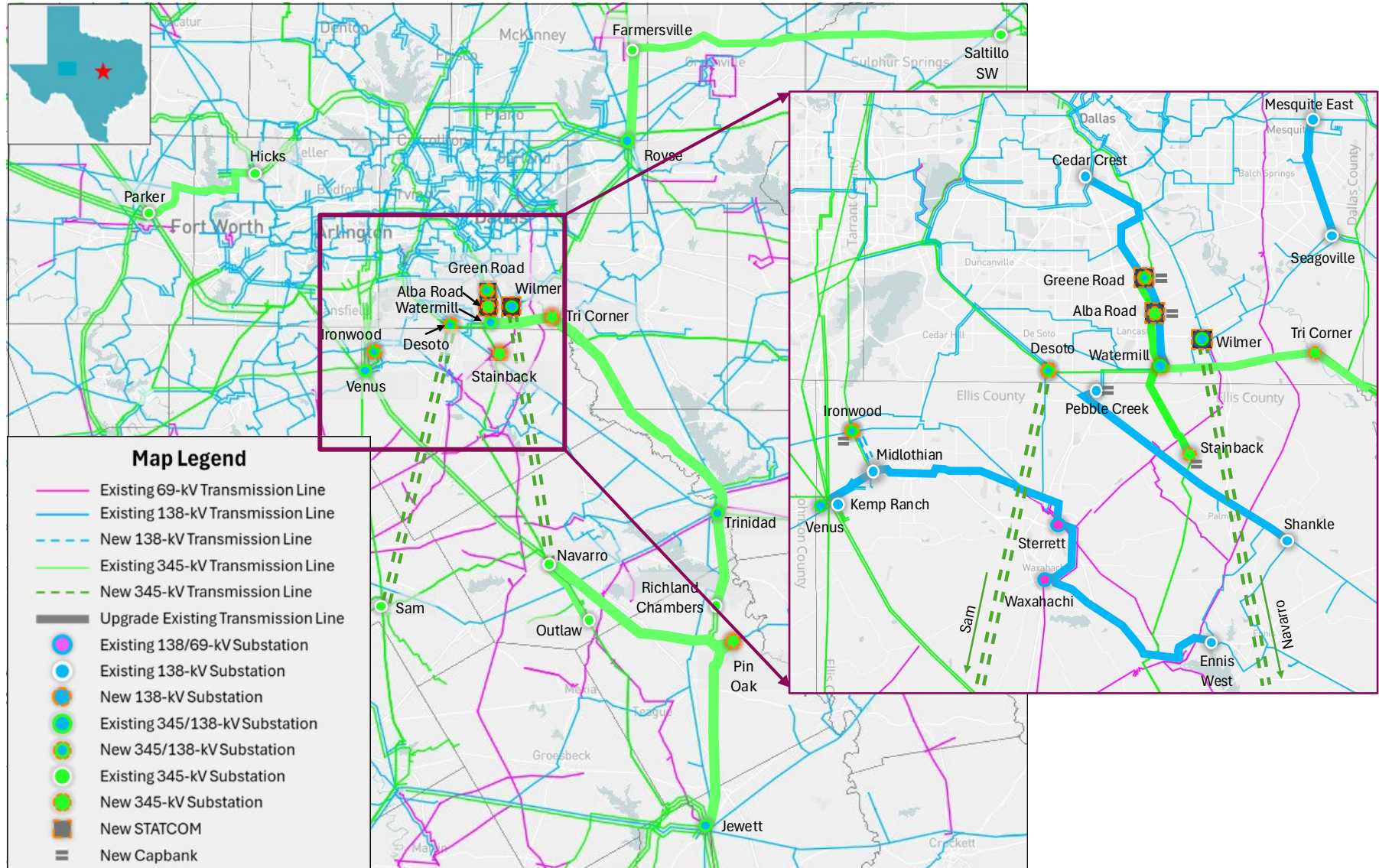
Option 2 – Modified Oncor Proposed Option Components Summary

- In addition to Option 1
 - Establish two new 138-kV substations
 - Goatheard Switch
 - Goatpad Switch
 - Construct six new 138-kV lines, approximately 14.4 circuit miles
 - Upgrade existing 138-kV lines, approximately 0.7 circuit miles
- Full project details in [Appendix E](#)

Option 3 – Modified Option 2 Components Summary

- In addition to Option 2
 - Construct two new 345-kV lines, approximately 218.0 circuit miles
 - Upgrade existing 345-kV lines, approximately 225.0 circuit miles
 - Upgrade existing switches
 - Four 345-kV switches
 - One 345/138-kV switch
- Full project details in [Appendix F](#)

Map of Option 3



Results of Reliability Assessments

- ERCOT conducted steady-state load flow analysis for the study base case according to the NERC Reliability Standard TPL-001-5.1 and ERCOT Planning Criteria to identify project need and to evaluate the proposed option and alternatives

Option	N-1		G-1*+N-1		X-1*+N-1		Unsolved Powerflow
	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	Thermal Violations	Voltage Violations	
1	11	3	3	None	2	None	None
2	2	None	4	None	2	None	None
3	None	None	None	None	None	None	None

* See [Appendix C](#) for list of G-1 generators and X-1 transformers tested

- Option 3 observed no reliability violations

Results of Maintenance Outage Evaluation

- ERCOT conducted maintenance outage analysis on all three options to compare relative performance of the options
 - Load levels the North Central and East Weather Zones were scaled down based on the historical non-summer peak data to 82% and 76%, respectively, in order to mimic the non-summer peak load condition
 - Based on the review of system topology of the area, ERCOT tested N-2 contingency combinations, and then tested all applicable contingency violations with system adjustments (N-1-1)
- Preliminary Finding shows no thermal overloads or voltage violations were observed in the N-1-1 analysis for Option 3

Option	Thermal Violation	Voltage Violation
Base Case	27	3
1	2	4
2	2	4
3	None	None

Long-Term Load-Serving Capability Assessment

- Assumptions
 - Adjusted load up in the study area, excluding Flexible Loads in the area
 - Adjusted conforming load down outside of the North Central and East Weather Zones to balance power
 - Based on N-1 contingency
- Preliminary Findings
 - Option 3 provides approximately 600 MW more incremental load-serving capability than Option 1 and Option 2

Option	Load-Serving Capability (~MW)
1	2305
2	2335
3	2904

Cost Estimate and Feasibility Assessment

- Transmission Service Providers (TSPs) performed feasibility assessments and provided cost estimates for the three options

Option	Cost Estimates (~\$B)	CCN Required (~miles)	Feasibility	Expected ISD
1	\$1.248	2.0	Yes	Oncor: May 2026 to December 2029
2	\$1.266	2.0	Yes	Oncor: May 2026 to December 2029 Brazos: December 2028
3	\$2.886	111.0	Yes	Oncor: May 2026 to April 2033 Brazos: December 2028 Lone Star: June 2030, April 2033

- Option 3 Cost Estimate includes
 - Approximately \$856.28 million for the upgrades identified in the 25RPG038 Oncor and Lone Star Navarro and Ellis Area Reliability Project
 - Approximately \$763.58 million for the upgrades identified as part of 25RPG040 and 25RPG042 Oncor Set 1 and Set 2 North and Central Texas Reliability Projects

Comparison of Options

	Option 1	Option 2	Option 3
Meets ERCOT and NERC Reliability Criteria	No	No	Yes
Improves Long-Term Load-Serving Capability (~MW)	Yes (2305)	Yes (2335)	Yes (2904)
Requires CCN (~miles)	Yes (2.0)	Yes (2.0)	Yes (56.5)
Cost Estimate* (~\$B)	\$1.248**	\$1.266	\$2.886
Feasible	Yes	Yes	Yes
Expected ISD	Oncor: May 2026 to December 2029	Oncor: May 2026 to December 2029 Brazos: December 2028	Oncor: May 2026 to April 2033 Brazos: December 2028 Lone Star: June 2030, April 2033

* Cost estimate was provided by the TSPs and includes the estimated capital cost with energized construction work.

** Cost estimate is different from initial project submittal

- Option 3 observed no reliability violations and met the project need
- Option 3 provides the best long-term load-serving capability

ERCOT Preferred Option

- Option 3 was selected as the ERCOT preferred option because it
 - Addresses the project need in the study area
 - Meets both ERCOT and NERC reliability criteria
 - Improves long-term load-serving capability for future load growth in the area
 - Improves operational flexibility
 - Observed no reliability issues

Generation Addition and Load Scaling Sensitivity Analyses

- Generation Addition Sensitivity Analysis
 - Per Planning Guide Section 3.1.3(4)(a), ERCOT performed a generation addition sensitivity by adding the new generation listed in [Appendix G](#) to the preferred option case. The additional 17 resources were modeled following the 2024 RTP methodology. ERCOT determined relevant generators do not impact the preferred option
- Load Scaling Sensitivity Analysis
 - Planning Guide Section 3.1.3(4)(b) requires an evaluation of the potential impact of load scaling on the criteria violations seen in this EIR. Starting 2024, ERCOT RTP adopted a new methodology of having one summer peak case for each study year with non-coincident peaks for each of the Weather Zones, which would eliminate the load scaling impact. The study case did not include load scaling as such load scaling sensitivity analysis is no longer needed

Additional Analyses

- Congestion analysis was performed for the preferred option using the 2025 RTP 2030 economic case
 - The ERCOT preferred option did not result in significant new congestion within the study area
- Subsynchronous Oscillations (SSO) Assessment was conducted for the preferred option per Nodal Protocol Section 3.22.1.3
 - ERCOT found no adverse SSO impacts to the existing and planned generation resources at the time of this study

Dynamic Stability Analysis

- ERCOT performed a dynamic stability analysis for the ERCOT's preferred option
- The study utilized the 2024 DWG 2026 Summer Peak dynamic case provided by Oncor, and the following were incorporated and evaluated in the analysis:
 - Large loads included in Oncor's RPG proposal within the study area
 - Critical contingencies provided by Oncor and additional contingencies near Large Load locations in the study area
 - Option 3 (ERCOT's preferred option)
- ERCOT observed no adverse dynamic stability impacts resulting from the implementation of Option 3 (ERCOT's preferred option)

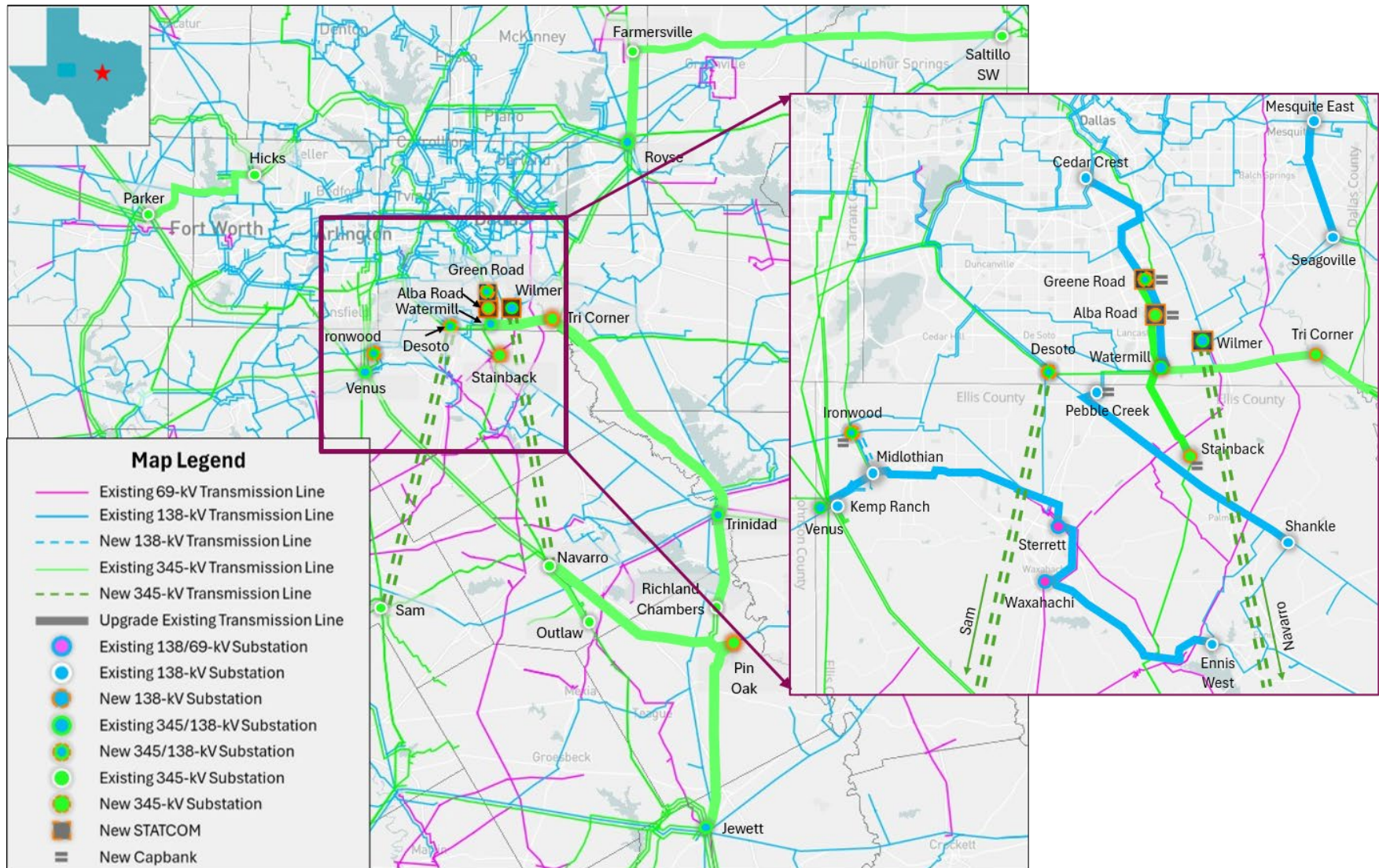
ERCOT Recommendation

- ERCOT recommends Option 3
 - Estimated Cost: approximately \$2.886 billion
 - Expected ISD:
 - For Oncor: between May 2026 to April 2033
 - For Brazos: December 2028
 - For Lone Star: June 2030 and April 2033
 - CCN filling will be required for the new transmission lines
 - From the existing Desoto Switch to the existing Sam Switch
 - From the existing Wilmer Switch to the existing Navarro Switch
 - From the existing Midlothian Tap to the new Ironwood Switch
 - The upgrades identified in Oncor and Lone Star Navarro and Ellis Area upgrade project (25RPG038) were included in its entirety as part of this project
 - A subset of upgrades identified in Oncor Set 1 and Set 2 North and Central Texas Reliability Projects (25RPG040 and 25RPG042) were included as part of this project

ERCOT Recommendation Components Summary

- Construct six new substations
 - Two new 345/138-kV switches
 - Two new 345-kV switches
 - Two new 138-kV substations
- Rebuild six existing substations
 - Five 345-kV switches
 - One 345/138-kV switches
- Construct twelve new 345-kV and 138-kV lines
 - Approximately 218.0 circuit miles of new 345-kV lines
 - Approximately 22.4 circuit miles of new 138-kV lines
- Upgrade of existing 345-kV and 138-kV lines
 - Approximately 394.8 circuit miles of existing 345-kV lines
 - Approximately 73.8 circuit miles of existing 138-kV lines
- Install nine reactive devices
 - Three ± 250 MVA 345-kV STATCOMs
 - Three 345-kV capacitor banks
 - Three 138-kV capacitor banks
- Full project details in [Appendix F](#)

Map of ERCOT Recommended Option



Next Steps and Tentative Timeline

- EIR report to be posted in the MIS in March 2026
- EIR recommendation to TAC in March 2026
- Seek ERCOT Board of Directors endorsement in April 2026

Thank you!



Stakeholder comments also welcomed through:

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Appendix A.1 – Transmission Projects Added

- List of recently approved RPG transmission projects added to study base case

RPG/TPIT No	Project Name	Tier	Project ISD	From County
24RPG016	Rand Area Loop Project	Tier 2	4/1/2027	Kaufman
24RPG017	Venus SW to Sam Switch 345-kV Line Project	Tier 1	5/1/2026	Ellis, Hill
24RPG022	Wilmer 345/138-kV Switch Project	Tier 1	5/1/2026	Dallas
24RPG025	Gunter 345/138-kV Switch Project	Tier 3	12/1/2025	Ellis, Hill

Appendix A.2 – Transmission Projects Removed

- List of placeholder transmission projects that are directly related to proposed project were removed from the study base case

Project ID	Project Name	County(s)
2024-E3	Big Brown SES West (3381) to Jewett (3391) 345-kV Line Upgrade and Substation Rebuilds	Henderson, Freestone
2024-E12	Trinidad SES (3124) to Richland Chambers (3134) 345-kV Line Upgrade	Freestone, Leon
2024-E18	Big Brown SES West (3381) to Jewett (3391) 345-kV Line Upgrade	Freestone, Leon
2024-NC13	Pebble Creek (2229) to Trumbull (221) to Gamma (12344) to Shankle Switch (12329) 138-kV Line Upgrades	Ellis
2024-NC14	Green Road (3069) to Ten Mile (2126) to Watermill (2429) to Reindeer (3065) 138-kV Line Upgrades	Dallas
2024-NC18	Ennis West Switch (2241) to Templeton (12320) to Waxahachie (2321) 138-kV Line Upgrades	Ellis, Waxahachie
2024-NC31	Royse Area 345-kV Line Upgrades and Substation Rebuilds	Collin, Hopkins, Rockwall, Fannin, Lamar
2024-NC63	Watermill Area 345-kV Line Additions and Reactive Support	Dallas, Navarro, Hill, Leon
2024-NC68	Batchler Road (2217) to Watermill (2427) 345-kV Line Upgrades	Dallas
2024-NC70	Miller Road (2632) 345/138-kV Substation Addition and 345-kV Lines Re-Termination	Dallas, Ellis
2024-NC85	Miller Road (2635) to Kemp Ranch Switch (2303) 138-kV double-circuit transmission line Upgrade	Ellis
2024-NC86	Greene Road (3063/3069) New 138-kV and 345-kV Line Additions and Substation Rebuilds	Dallas

Appendix A.2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

Project ID	Project Name	County(s)
2024-E2	Shamburger (3103) to Elkton (3105) 345-kV Double Circuit Line Addition	Smith
2024-N11	Alla Hubbard (1757) to Epco POI (12468) 138-kV Line Upgrade	Collin, Grayson
2024-NC01	Glen Heights (217) to Sterett Road (2237) 138-kV Line Upgrade	Ellis
2024-NC04	Railport (442) Area 138-kV Rebuild	Ellis, Johnson
2024-NC06	Miller (824) to Newman (849) 138-kV Line Upgrade	Dallas
2024-NC07	Walnut (823) to Naaman (825) 138-kV Line Upgrade	Dallas
2024-NC08	Centerville (810) to Miller (824) 138-kV Line Upgrade	Dallas
2024-NC09	Walnut (823) to Newman (849) 138-kV Line Upgrade	Dallas
2024-NC10	Glen Heights (217) to Desoto Switch 8 (2424) 138-kV Line Upgrade	Ellis, Dallas
2024-NC11	Watermill (2427) to Tricorner (2432) to Trinidad (3123/3124) 345-kV Line Upgrades	Dallas, Henderson
2024-NC15	Waxahachie (2321) to Waxahachie North (2320) 138-kV Line Upgrade and Waxahachie (2335) to Waxahachie OCF (2343) 69-kV Line Upgrades	Ellis
2024-NC19	Climax TNP (37280) to Bridge Point RC (37245) 69-kV Line Upgrade	Collin
2024-NC21	Carrollton Northwest (2363) to South TNP (37100) to TI TNP (37080) 138-kV Line Upgrades	Dallas, Denton
2024-NC22	Waxahachie Pump 2 (2315) to Waxahachie North (2320) 138-kV Line Upgrade	Ellis

Appendix A.2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

Project ID	Project Name	County(s)
2024-NC25	LTV Sub (2259) to Grand Prairie (2262) 138-kV Line Upgrade	Dallas
2024-NC27	Navarro (3478) to Haney (213) to Payne Battle Creek POI (888876) to Hubbard (3515) 138-kV Line Upgrades	Navarro, Hill
2024-NC28	Olympus Switch (3723) to Montfort Switch (3454) to Chatfield (200) 138-kV Line Upgrade	Henderson, Navarro
2024-NC29	Allen Switch (2514) to Pine Forest POI (888854) 345-kV Line Upgrade	Collin, Hopkins
2024-NC34	Trinidad SES (3127) to Nipak Tap (3260) to Mankin Switch (3265) 138-kV Line Upgrade	Henderson, Navarro
2024-NC35	Firewheel (821) to Wylie Switch (833) 138-kV Line Upgrade	Dallas, Collin
2024-NC38	Seagoville Area Upgrades	Dallas
2024-NC41	Ben Davis (968) to GOBLN_8 (2497) 138-kV Line Upgrade	Dallas
2024-NC44	Cedar Creek 138-kV Area Line Upgrades	Henderson
2024-NC51	Waxahachie-Sterrett Area Upgrades	Ellis
2024-NC59	Lavon Switch 345/138-kV Switch Substation Expansion and Lavon Switch (2475) to Allen Switch (2511) 138-kV Line Addition	Collin, Rockwall
2024-NC64	Calmont Switch (1955) to Benbrook Switch (1874) 138-kV Line Upgrade	Tarrant
2024-NC66	Tri Corner (2432) to Forney Switch (2437) 345-kV Line Upgrade	Dallas, Kaufman

Appendix A.2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

Project ID	Project Name	County(s)
2024-NC67	Elkton (3105) to Tri Corner (2432) 345-kV Line Additions and Upgrades	Dallas, Kaufman, Smith, Van Zandt
2024-NC71	Mitchell Bend Switch (1895) to Carmichael Bend Switch (2285) to Benbrook Switch (1873) 345-kV Line Upgrades	Tarrant, Hood, Grayson
2024-NC74	Midlothian TXI (2307) to Waxahachie Northwest (2309) 138-kV Line Upgrades	Ellis
2024-NC75	Primrose Fort Worth (2196) to Rocky Creek (1881) to Primrose Fort Worth (2197) 138-kV Line Upgrades	Tarrant
2024-NC76	Gunter 345/138-kV Switch Addition and Gunter (2236) to Collin (2370) 138-kV Line Addition	Cooke, Denton, Collin, Grayson
2024-NC77	Skyview 345/138-kV Switch Addition and Area Upgrade	Dallas, Tarrant
2024-NC79	Upgrade Benbrook Switch 345/138-kV (1869/18741) and (1870/18751) Transformers	Tarrant
2024-NC80	Desoto Switch 345/138-kV Transformer Addition (2431/2424) and Desoto Switch (2424) to Loop9 (2848) 138-kV Line Addition	Dallas
2024-NC82	Rocky Creek 345/138-kV Transformer Addition (1880/1881)	Tarrant
2024-NC83	Hempill (2164) to Mistletoe Heights (2173) 138-kV Line Upgrade	Tarrant
2024-NC84	Wedgewood Switch (2184) to Bryant Irvin (2182) 138-kV Line Upgrade	Tarrant

Appendix B – New Generation Projects Added

- List of new generation, that met the Planning Guide 6.9(1) condition, added to the study base case

GINR	Project Name	Fuel	Projected COD	Capacity (~MW)	County
21INR0359	Hickerson Solar	SOL	3/1/2026	316.3	Bosque
22INR0437	TORMES SOLAR	SOL	3/31/2027	382.1	Navarro
24INR0126	High Noon Storage	OTH	12/1/2027	94.0	Hill
24INR0188	Tehuacana Creek Solar SLF	SOL	3/10/2027	505.5	Navarro
24INR0189	Tehuacana Creek BESS SLF	OTH	3/10/2027	419.0	Navarro
24INR0355	Anatole Renewable Energy Storage	OTH	1/11/2026	207.8	Henderson
24INR0364	Pitts Dudik II	SOL	1/29/2026	30.2	Hill
24INR0498	Fort Watt Storage	OTH	4/20/2027	205.4	Tarrant
24INR0631	Radian Storage SLF	OTH	4/22/2025	160.3	Brown
25INR0018	Yellow Cat Wind	WIN	9/30/2026	301.2	Navarro
25INR0046	Blue Skies BESS	OTH	12/31/2027	306.3	Hill
25INR0231	Apache Hill BESS	OTH	11/15/2026	200.9	Hood
25INR0391	Purple Sage BESS 1	OTH	5/30/2027	156.0	Collin
25INR0392	Purple Sage BESS 2	OTH	5/30/2027	156.0	Collin
26INR0543	Three Canes Solar SLF	SOL	12/31/2026	333.0	Navarro

Appendix C – G-1 Generators and X-1 Transformers

G-1 Generators	X-1 Transformers
Comanche Peak	Desoto Switch 345/138-kV
Forney Combine Cycle Train	Monticello SES 345/138-kV
Martain Lake	Sargent Road Switch 345/138-kV
Mt. Creek SES	Seagoville Switch 345/138-kV
Midlothian	Trinidad SES Switch 345/138-kV
	Venus Kemp Ranch Switch 345/138-kV
	Watermill Switch 345/138-kV
	West Levee Switch 345/138-kV
	Wilmer Switch 345/138-kV

Appendix D – Option 1 – Oncor Proposed Option Details

- Construct a new Greene Road 345/138-kV Switch (SW) by installing fourteen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement:
 - Install two new 345/138-kV autotransformers with a normal rating of 700 MVA and an emergency rating of 750 MVA;
 - Re-terminate the existing Watermill SW to Sargent Road SW/West Levee SW 345-kV double-circuit transmission line into the new Greene Road 345-kV SW, which creates the new Watermill SW to Greene Road SW and the new Greene Road SW to Sargent Road SW/West Levee SW 345-kV double-circuit transmission lines;
 - Re-terminate the existing Wilson SW to Cedar Hill SW/Cedar Crest SW 138-kV double-circuit transmission line into the new Greene Road 138-kV SW, which creates the new Wilson SW to Greene Road SW and the new Greene Road SW to Cedar Hill SW/Cedar Crest SW 138-kV double-circuit transmission line;
- Rebuild the new Greene Road SW to Watermill SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.6 miles;
- Rebuild the new Wilson SW to Greene Road SW 138-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 764 MVA, approximately 2.0 miles;
- Rebuild the new Greene Road SW to Cedar Crest 138-kV single-circuit transmission line on double-circuit structures with one circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 10.9 miles;

Appendix D – Option 1 – Oncor Proposed Option Details (cont.)

- Construct a new Alba Road 345-kV SW by installing eleven 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement:
 - Re-terminate the new Watermill SW to Greene Road SW 345-kV double-circuit transmission line into the new Alba Road 345-kV SW, which creates the new Watermill SW to Alba Road SW and the new Alba Road SW to Greene Road SW 345-kV double-circuit transmission lines;
- Construct a new Stainback 345-kV SW by installing fourteen 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement:
 - Re-terminate the existing Watermill SW to Elrod SW/Big Onion SW 345-kV double-circuit transmission line into the new Stainback 345-kV SW, which creates the new Watermill SW to Stainback SW and the new Stainback SW to Elrod SW/Big Onion SW 345-kV double-circuit transmission lines;
- Rebuild the new Watermill SW to Stainback SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.0 miles;
- Rebuild the existing Watermill SW to Wilson SW 138-kV double-circuit transmission line on double-circuit structures with both circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 0.8 miles;

Appendix D – Option 1 – Oncor Proposed Option Details (cont.)

- Construct a new Ironwood 345/138-kV SW by installing seventeen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement:
 - Install two 345/138-kV autotransformers with normal and emergency ratings of 700 MVA and 750 MVA, respectively;
 - Re-terminate the existing Liggett SW to Endeavor SW 345-kV Line at the new Ironwood 345-kV SW, which creates the new Liggett SW to Venus SW (north bus)/Ironwood SW 345-kV double-circuit transmission line;
 - Disconnect the existing Endeavor SW to Venus SW (south bus) and Midlothian ANP #2 to Venus SW (south bus) 345-kV Lines from Venus SW (south bus) and connect the Midlothian ANP to Endeavor 345-kV SW. This will create Midlothian ANP #1 to Venus SW (north bus) and Midlothian ANP #2 to Endeavor SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Timberview SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Everman SW to Venus SW (north bus) and the new Timberview SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Cedar Hill SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Sherry SW to Venus SW (north bus) and the new Cedar Hill SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Sam SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Fort Smith SW to Venus SW (north bus) and the new Sam SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Navarro SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Navarro SW to Venus SW (north bus)/Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Cottonwood Creek 345/138-kV Autotransformer #2 at the north bus of Venus 345-kV SW by installing one 345-kV, 5,000 A breaker;

Appendix D – Option 1 – Oncor Proposed Option Details (cont.)

- Loop the existing Kemp Ranch SW to Sardis SW/Soap Creek 138-kV double-circuit transmission line into the new Ironwood 138-kV SW by disconnecting the double-circuit line at structure #1/2 (Midlothian Tap) and constructing four circuits from Midlothian Tap to the new Ironwood 138-kV SW on separate structures, with normal and emergency ratings of at least 764 MVA, approximately 2.0 miles each circuit;
- Rebuild the two Kemp Ranch SW to Midlothian Tap 138-kV single-circuit transmission line sections using two separate structures, with normal and emergency ratings of at least 764 MVA, approximately 0.5 miles each circuit;
- Rebuild the existing Sterrett SW to Midlothian TXI 138-kV single-circuit transmission line sections, with normal and emergency ratings of at least 764 MVA, approximately 11.8 miles;
- Rebuild the existing Ennis West SW to Sterrett SW 138-kV single-circuit transmission line, with normal and emergency ratings of at least 614 MVA, approximately 21.0 miles;

Appendix D – Option 1 – Oncor Proposed Option Details (cont.)

- Rebuild the existing Big Brown 345-kV SW by installing twelve 345-kV, 5,000 A breakers in a breaker-and-a-half bus arrangement. Upon completion, Big Brown SW will be renamed as Pin Oak SW:
 - Re-terminate the existing Big Brown SW to Jewett SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW to Jewett SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown SW to Navarro SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW – Navarro SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown SW to Richland Chamber SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW to Richland Chambers SW 345-kV double-circuit transmission line;
- Rebuild the new Jewett SW to Pin Oak SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 32.8 miles;
- Rebuild the existing Richland Chambers SW to Trinidad SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place with normal and emergency ratings of at least 1,792 MVA and a with conductor rating of at least 2,987 MVA, approximately 18.7 miles;

Appendix D – Option 1 – Oncor Proposed Option Details (cont.)

- Rebuild the existing Parker SW to Hicks SW 345-kV transmission line, with normal and emergency ratings of at least 1,912 MVA and with a conductor rating of at least 2,987 MVA, approximately 23.0 miles;
- Rebuild the existing Pebble Creek SW to Shankle SW 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 15.5 miles;
- Rebuild the existing Mesquite East SW to Seagoville SW 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 7.4 miles;
- Rebuild the existing Farmersville SW to Royse SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 15.3 miles;
- Install one +250/-250 MVar Grid-forming STATCOM at each of the following:
 - Alba Road 345-kVSW;
 - Greene Road 345-kVSW;
 - Wilmer 345kV SW;

Appendix D – Option 1 – Oncor Proposed Option Details (cont.)

- Install 240 MVar 345-kV capacitor banks (3-80 MVar each):
 - One at Greene Road 345-kV SW;
 - Two at Alba Road 345-kV SW;
 - Two at Stainback 345-kV SW;
- Install 110.4 MVar 138-kV capacitor banks (3-36.8 MVar each) at:
 - One at Greene Road 138-kVSW;
 - Two at Ironwood 138-kVSW;
 - Two at Pebble Creek 138-kVSW;
- For terminal equipment:
 - The existing 345 kV terminal equipment, ensure they meet or exceed a rating of 3000 A (1792 MVA);
 - The new 345-kV terminal equipment, ensure they meet or exceed a rating of 5,000 A if the station is 5,000 A capable. Otherwise ensure the new 345-kV terminal equipment meets or exceeds a rating of 3,200 A; and
 - The 138-kV terminal equipment, ensure they meet or exceed a rating of 3,000 A.

Appendix E – Option 2 (Addition to Option 1)

Details

- Construct a new Greene Road 345/138-kV Switch (SW) by installing fourteen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement:
 - Install two new 345/138-kV autotransformers with a normal rating of 700 MVA and an emergency rating of 750 MVA;
 - Re-terminate the existing Watermill SW to Sargent Road SW/West Levee SW 345-kV double-circuit transmission line into the new Greene Road 345-kV SW, which creates the new Watermill SW to Greene Road SW and the new Greene Road SW to Sargent Road SW/West Levee SW 345-kV double-circuit transmission lines;
 - Re-terminate the existing Wilson SW to Cedar Hill SW/Cedar Crest SW 138-kV double-circuit transmission line into the new Greene Road 138-kV SW, which creates the new Wilson SW to Greene Road SW and the new Greene Road SW to Cedar Hill SW/Cedar Crest SW 138-kV double-circuit transmission line;
- Rebuild the new Greene Road SW to Watermill SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.6 miles;
- Rebuild the new Wilson SW to Greene Road SW 138-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 764 MVA, approximately 2.0 miles;
- Rebuild the new Greene Road SW to Cedar Crest 138-kV single-circuit transmission line on double-circuit structures with one circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 10.9 miles;

Appendix E – Option 2 (Addition to Option 1)

Details (cont.)

- Construct a new Alba Road 345-kV SW by installing eleven 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement:
 - Re-terminate the new Watermill SW to Greene Road SW 345-kV double-circuit transmission line into the new Alba Road 345-kV SW, which creates the new Watermill SW to Alba Road SW and the new Alba Road SW to Greene Road SW 345-kV double-circuit transmission lines;
- Construct a new Stainback 345-kV SW by installing fourteen 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement:
 - Re-terminate the existing Watermill SW to Elrod SW/Big Onion SW 345-kV double-circuit transmission line into the new Stainback 345-kV SW, which creates the new Watermill SW to Stainback SW and the new Stainback SW to Elrod SW/Big Onion SW 345-kV double-circuit transmission lines;
- Rebuild the new Watermill SW to Stainback SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.0 miles;
- Rebuild the existing Watermill SW to Wilson SW 138-kV double-circuit transmission line on double-circuit structures with both circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 0.8 miles;

Appendix E – Option 2 (Addition to Option 1)

Details (cont.)

- Construct a new Ironwood 345/138-kV SW by installing seventeen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement:
 - Install two 345/138-kV autotransformers with normal and emergency ratings of 700 MVA and 750 MVA, respectively;
 - Re-terminate the existing Liggett SW to Endeavor SW 345-kV Line at the new Ironwood 345-kV SW, which creates the new Liggett SW to Venus SW (north bus)/Ironwood SW 345-kV double-circuit transmission line;
 - Disconnect the existing Endeavor SW to Venus SW (south bus) and Midlothian ANP #2 to Venus SW (south bus) 345-kV Lines from Venus SW (south bus) and connect the Midlothian ANP to Endeavor 345-kV SW. This will create Midlothian ANP #1 to Venus SW (north bus) and Midlothian ANP #2 to Endeavor SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Timberview SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Everman SW to Venus SW (north bus) and the new Timberview SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Cedar Hill SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Sherry SW to Venus SW (north bus) and the new Cedar Hill SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Sam SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Fort Smith SW to Venus SW (north bus) and the new Sam SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Navarro SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Navarro SW to Venus SW (north bus)/Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Cottonwood Creek 345/138-kV Autotransformer #2 at the north bus of Venus 345-kV SW by installing one 345-kV, 5,000 A breaker;

Appendix E – Option 2 (Addition to Option 1) Details (cont.)

- Loop the existing Kemp Ranch SW to Sardis SW/Soap Creek 138-kV double-circuit transmission line into the new Ironwood 138-kV SW by disconnecting the double-circuit line at structure #1/2 (Midlothian Tap) and constructing four circuits from Midlothian Tap to the new Ironwood 138-kV SW on separate structures, with normal and emergency ratings of at least 764 MVA, approximately 2.0 miles each circuit;
- Rebuild the two Kemp Ranch SW to Midlothian Tap 138-kV single-circuit transmission line sections using two separate structures, with normal and emergency ratings of at least 764 MVA, approximately 0.5 miles each circuit;
- Rebuild the existing Sterrett SW to Midlothian TXI 138-kV single-circuit transmission line sections, with normal and emergency ratings of at least 764 MVA, approximately 11.8 miles;
- Rebuild the existing Ennis West SW to Sterrett SW 138-kV single-circuit transmission line, with normal and emergency ratings of at least 614 MVA, approximately 21.0 miles;

Appendix E – Option 2 (Addition to Option 1)

Details (cont.)

- Rebuild the existing Big Brown 345-kV SW by installing twelve 345-kV, 5,000 A breakers in a breaker-and-a-half bus arrangement. Upon completion, Big Brown SW will be renamed as Pin Oak SW:
 - Re-terminate the existing Big Brown SW to Jewett SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW to Jewett SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown SW to Navarro SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW – Navarro SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown SW to Richland Chamber SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW to Richland Chambers SW 345-kV double-circuit transmission line;
- Rebuild the new Jewett SW to Pin Oak SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 32.8 miles;
- Rebuild the existing Richland Chambers SW to Trinidad SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place with normal and emergency ratings of at least 1,792 MVA and a with conductor rating of at least 2,987 MVA, approximately 18.7 miles;

Appendix E – Option 2 (Addition to Option 1) Details (cont.)

- Rebuild the existing Parker SW to Hicks SW 345-kV transmission line, with normal and emergency ratings of at least 1,912 MVA and with a conductor rating of at least 2,987 MVA, approximately 23.0 miles;
- Rebuild the existing Pebble Creek SW to Shankle SW 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 15.5 miles;
- Rebuild the existing Mesquite East SW to Seagoville SW 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 7.4 miles;
- Rebuild the existing Farmersville SW to Royse SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 15.3 miles;
- Install one +250/-250 MVar Grid-forming STATCOM at each of the following:
 - Alba Road 345-kVSW;
 - Greene Road 345-kVSW;
 - Wilmer 345kV SW;

Appendix E – Option 2 (Addition to Option 1)

Details (cont.)

- Install 240 MVar 345-kV capacitor banks (3-80 MVar each):
 - One at Greene Road 345-kV SW;
 - Two at Alba Road 345-kV SW;
 - Two at Stainback 345-kV SW;
- Install 110.4 MVar 138-kV capacitor banks (3-36.8 MVar each) at:
 - One at Greene Road 138-kVSW;
 - Two at Ironwood 138-kVSW;
 - Two at Pebble Creek 138-kVSW;
- For terminal equipment:
 - The existing 345 kV terminal equipment, ensure they meet or exceed a rating of 3000 A (1792 MVA);
 - The new 345-kV terminal equipment, ensure they meet or exceed a rating of 5,000 A if the station is 5,000 A capable. Otherwise ensure the new 345-kV terminal equipment meets or exceeds a rating of 3,200 A;
 - The 138-kV terminal equipment, ensure they meet or exceed a rating of 3,000 A;

Appendix E – Option 2 (Addition to Option 1)

Details (cont.)

- Construct a new Goatpad 138-kV Substation:
 - Install a new Goatpad to Padera 138-kV single-circuit transmission line on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 287 MVA, approximately 2.6 miles;
 - Install a new Goatpad to Midlothian 138-kV single-circuit transmission line on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 287 MVA, approximately 3.0 miles;
 - Install a new Goatpad to Goatheard 138-kV single-circuit transmission line on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 524 MVA, approximately 0.1 miles; and
- Construct a new Goatheard 138-kV Substation:
 - Re-terminate the existing Cedar Hill SW to Saint Paul SW 138-kV single-circuit transmission line into the new Goatheard 138-kV SW, which creates the new Goatheard SW to Cedar Hill SW and the new Goatheard SW to Saint Paul SW 138-kV single-circuit transmission line, and rebuild on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 237 MVA, approximately 0.7 miles.

Appendix F – Option 3 (Addition to Option 2)

Details

- Construct a new Greene Road 345/138-kV Switch (SW) by installing fourteen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement:
 - Install two new 345/138-kV autotransformers with a normal rating of 700 MVA and an emergency rating of 750 MVA;
 - Re-terminate the existing Watermill SW to Sargent Road SW/West Levee SW 345-kV double-circuit transmission line into the new Greene Road 345-kV SW, which creates the new Watermill SW to Greene Road SW and the new Greene Road SW to Sargent Road SW/West Levee SW 345-kV double-circuit transmission lines;
 - Re-terminate the existing Wilson SW to Cedar Hill SW/Cedar Crest SW 138-kV double-circuit transmission line into the new Greene Road 138-kV SW, which creates the new Wilson SW to Greene Road SW and the new Greene Road SW to Cedar Hill SW/Cedar Crest SW 138-kV double-circuit transmission line;
- Rebuild the new Greene Road SW to Watermill SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.6 miles;
- Rebuild the new Wilson SW to Greene Road SW 138-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 764 MVA, approximately 2.0 miles;
- Rebuild the new Greene Road SW to Cedar Crest 138-kV single-circuit transmission line on double-circuit structures with one circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 10.9 miles;

Appendix F – Option 3 (Addition to Option 2)

Details (cont.)

- Construct a new Alba Road 345-kV SW by installing eleven 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement:
 - Re-terminate the new Watermill SW to Greene Road SW 345-kV double-circuit transmission line into the new Alba Road 345-kV SW, which creates the new Watermill SW to Alba Road SW and the new Alba Road SW to Greene Road SW 345-kV double-circuit transmission lines;
- Construct a new Stainback 345-kV SW by installing fourteen 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement:
 - Re-terminate the existing Watermill SW to Elrod SW/Big Onion SW 345-kV double-circuit transmission line into the new Stainback 345-kV SW, which creates the new Watermill SW to Stainback SW and the new Stainback SW to Elrod SW/Big Onion SW 345-kV double-circuit transmission lines;
- Rebuild the new Watermill SW to Stainback SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.0 miles;
- Rebuild the existing Watermill SW to Wilson SW 138-kV double-circuit transmission line on double-circuit structures with both circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 0.8 miles;

Appendix F – Option 3 (Addition to Option 2)

Details (cont.)

- Construct a new Ironwood 345/138-kV SW by installing seventeen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement:
 - Install two 345/138-kV autotransformers with normal and emergency ratings of 700 MVA and 750 MVA, respectively;
 - Re-terminate the existing Liggett SW to Endeavor SW 345-kV Line at the new Ironwood 345-kV SW, which creates the new Liggett SW to Venus SW (north bus)/Ironwood SW 345-kV double-circuit transmission line;
 - Disconnect the existing Endeavor SW to Venus SW (south bus) and Midlothian ANP #2 to Venus SW (south bus) 345-kV Lines from Venus SW (south bus) and connect the Midlothian ANP to Endeavor 345-kV SW. This will create Midlothian ANP #1 to Venus SW (north bus) and Midlothian ANP #2 to Endeavor SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Timberview SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Everman SW to Venus SW (north bus) and the new Timberview SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Cedar Hill SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Sherry SW to Venus SW (north bus) and the new Cedar Hill SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Sam SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Fort Smith SW to Venus SW (north bus) and the new Sam SW to Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Navarro SW to Venus SW (south bus) at the new Ironwood 345-kV SW, which creates the new Navarro SW to Venus SW (north bus)/Ironwood SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Cottonwood Creek 345/138-kV Autotransformer #2 at the north bus of Venus 345-kV SW by installing one 345-kV, 5,000 A breaker;

Appendix F – Option 3 (Addition to Option 2) Details (cont.)

- Loop the existing Kemp Ranch SW to Sardis SW/Soap Creek 138-kV double-circuit transmission line into the new Ironwood 138-kV SW by disconnecting the double-circuit line at structure #1/2 (Midlothian Tap) and constructing four circuits from Midlothian Tap to the new Ironwood 138-kV SW on separate structures, with normal and emergency ratings of at least 764 MVA, approximately 2.0 miles each circuit;
- Rebuild the two Kemp Ranch SW to Midlothian Tap 138-kV single-circuit transmission line sections using two separate structures, with normal and emergency ratings of at least 764 MVA, approximately 0.5 miles each circuit;
- Rebuild the existing Sterrett SW to Midlothian TXI 138-kV single-circuit transmission line sections, with normal and emergency ratings of at least 764 MVA, approximately 11.8 miles;
- Rebuild the existing Ennis West SW to Sterrett SW 138-kV single-circuit transmission line, with normal and emergency ratings of at least 614 MVA, approximately 21.0 miles;

Appendix F – Option 3 (Addition to Option 2) Details (cont.)

- Rebuild the existing Big Brown 345-kV SW by installing twelve 345-kV, 5,000 A breakers in a breaker-and-a-half bus arrangement. Upon completion, Big Brown SW will be renamed as Pin Oak SW:
 - Re-terminate the existing Big Brown SW to Jewett SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW to Jewett SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown SW to Navarro SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW – Navarro SW 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown SW to Richland Chamber SW 345-kV double-circuit transmission line at the new Pin Oak 345-kV SW, which creates the new Pin Oak SW to Richland Chambers SW 345-kV double-circuit transmission line;
- Rebuild the new Jewett SW to Pin Oak SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 32.8 miles;
- Rebuild the existing Richland Chambers SW to Trinidad SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place with normal and emergency ratings of at least 1,792 MVA and a with conductor rating of at least 2,987 MVA, approximately 18.7 miles;

Appendix F – Option 3 (Addition to Option 2) Details (cont.)

- Rebuild the existing Parker SW to Hicks SW 345-kV transmission line, with normal and emergency ratings of at least 1,912 MVA and with a conductor rating of at least 2,987 MVA, approximately 23.0 miles;
- Rebuild the existing Pebble Creek SW to Shankle SW 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 15.5 miles;
- Rebuild the existing Mesquite East SW to Seagoville SW 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 7.4 miles;
- Rebuild the existing Farmersville SW to Royse SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 15.3 miles;
- Install one +250/-250 MVar Grid-forming STATCOM at each of the following:
 - Alba Road 345-kVSW;
 - Greene Road 345-kVSW;
 - Wilmer 345kV SW;

Appendix F – Option 3 (Addition to Option 2)

Details (cont.)

- Install 240 MVar 345-kV capacitor banks (3-80 MVar each):
 - One at Greene Road 345-kV SW;
 - Two at Alba Road 345-kV SW;
 - Two at Stainback 345-kV SW;
- Install 110.4 MVar 138-kV capacitor banks (3-36.8 MVar each) at:
 - One at Greene Road 138-kVSW;
 - Two at Ironwood 138-kVSW;
 - Two at Pebble Creek 138-kVSW;
- For terminal equipment:
 - The existing 345 kV terminal equipment, ensure they meet or exceed a rating of 3000 A (1792 MVA);
 - The new 345-kV terminal equipment, ensure they meet or exceed a rating of 5,000 A if the station is 5,000 A capable. Otherwise ensure the new 345-kV terminal equipment meets or exceeds a rating of 3,200 A;
 - The 138-kV terminal equipment, ensure they meet or exceed a rating of 3,000 A;

Appendix F – Option 3 (Addition to Option 2)

Details (cont.)

- Construct a new Goatpad 138-kV Substation:
 - Install a new Goatpad to Padera 138-kV single-circuit transmission line on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 287 MVA, approximately 2.6 miles;
 - Install a new Goatpad to Midlothian 138-kV single-circuit transmission line on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 287 MVA, approximately 3.0 miles;
 - Install a new Goatpad to Goatheard 138-kV single-circuit transmission line on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 524 MVA, approximately 0.1 miles;
- Construct a new Goatheard 138-kV Substation:
 - Re-terminate the existing Cedar Hill SW to Saint Paul SW 138-kV single-circuit transmission line into the new Goatheard 138-kV SW, which creates the new Goatheard SW to Cedar Hill SW and the new Goatheard SW to Saint Paul SW 138-kV single-circuit transmission line, and rebuild on double-circuit structures with only one circuits in place, with normal and emergency ratings of at least 237 MVA, approximately 0.7 miles;

Appendix F – Option 3 (Addition to Option 2) Details (cont.)

- Upgrade the existing Navarro SW to the new Pin Oak (Big Brown) SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 27.6 miles;
- Upgrade the existing Navarro SW to Outlaw SW 345-kV double-circuit transmission line on double-circuit structures with both one circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 5.6 miles;
- Rebuild the existing Desoto 345/138-kV SW:
 - Install five 345-kV circuit breakers in a breaker-and-a-half configuration with ratings of 5,000 A;
 - Install nine 138-kV circuit breakers in a breaker-and-a-half configuration with ratings of 3,200 A;
- Construct a new Desoto SW to Sam SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 56.0 miles/circuit:
 - Install four 345-kV circuit breakers at the existing Desoto SW 345-kV substation with ratings of at least 5000 A;
 - Install four 345-kV circuit breakers at the existing Sam SW 345-kV substation with ratings of at least 5000 A;

Appendix F – Option 3 (Addition to Option 2)

Details (cont.)

- Construct a new Wilmer SW to Navarro SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 53.0 miles/circuit:
 - Install three 345-kV circuit breakers at the existing Wilmer SW 345-kV substation with ratings of at least 5000 A;
 - Install four 345-kV circuit breakers at the existing Navarro SW 345-kV substation with ratings of at least 5000 A;
- Rebuild the existing Tri Corner 345-kV SW:
 - Install twelve 345-kV circuit breakers in a breaker-and-a-half configuration with ratings of 5,000 A;
 - Ensure all terminal equipment meet or exceed 5,000A;
- Upgrade the existing Watermill SW to Tri Corner SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,912 MVA, approximately 11.5 miles/circuit:
 - Ensure all terminal equipment meet or exceed 5,000A;
- Upgrade the existing Tri Corner SW to Trinidad SW 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,912 MVA, approximately 40.5 miles/circuit:
 - Ensure all terminal equipment meet or exceed 5,000A; and
- Upgrade the existing Farmerville SW to Saltillo SW 345-kV single-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,912 MVA, approximately 59.9 miles/circuit:
 - Ensure all terminal equipment meet or exceed 3,200A.

Appendix G – New Generation Projects Added for Generation Addition Sensitivity Analysis

- List of new generation, that met the Planning Guide 6.9(1) condition, added to the study case for the Generation Addition Sensitivity Analysis

GINR	Project Name	Fuel	Projected COD	Max Capacity (~MW)	County
21INR0368	Eliza Solar	SOL	12/18/2025	151.7	Kaufman
22INR0220	Lamkin Solar	SOL	8/8/2027	101.5	Comanche
22INR0261	Dorado Solar	SOL	2/16/2026	401.4	Callahan
23INR0538	Roadrunner Crossing BESS SLF	OTH	2/13/2026	0.0	Eastland
24INR0136	Eagle Springs Storage	OTH	12/31/2026	33.1	Delta
24INR0137	Eagle Springs Solar	SOL	12/31/2026	77.2	Delta
24INR0181	Bynum Solar Project	SOL	2/6/2026	56.0	Coryell
25INR0101	Drake BESS	OTH	7/14/2026	257.3	Collin
25INR0122	Vial BESS	OTH	9/15/2027	135.2	Hill
25INR0166	Padrino Solar	SOL	7/30/2026	201.4	Hill
25INR0199	Bonham Solar SLF	SOL	4/6/2027	138.4	Limestone
25INR0229	OCI Cobb Creek Solar	SOL	12/31/2027	203.1	Hill
25INR0233	OCI Cobb Creek ESS	OTH	12/31/2027	201.6	Hill
25INR0281	Cosper Solar	SOL	11/12/2027	148.2	Bell
26INR0241	Sol Marina Energy Center	SOL	10/29/2027	175.3	Ellis
26INR0242	Sol Marina Energy Center BESS	OTH	10/29/2027	57.2	Ellis
26INR0380	Pepper Solar Farm	SOL	9/20/2027	120.9	McLennan